Band-1 Hadamard order 12240 – Classical Paley-I (q=12239); IN SPK via Paley-I; Gap 2052 Still OPEN

H_12240 Paley-I PASS; Paley-II FAIL/SKIP (q=6119=29*211 NOT prime; ===3 mod 4 needs ===1); Kronecker FAIL/SKIP (4*3060; 3060===0 mod 4 order-admissible; factors 2^2*3^2*5*17; not_in_seeds / no H_3060); Miyamoto digression skipped; gap 2052 OPEN; conjecture OPEN; GATE-04 LOCKED

C.M. Baird; ZoraASI mission-control Band-1 dash

2026-10-03

Document class. Scientific computational / construction certificate (Band-1).

Program. Open-mathematics collapse list * Human+AI attack order * ZoraASI mission-control.

Parents. 2026-10-03_Band1_Hadamard_12228_PaleyI; 2026-10-03_Band1_Hadamard_12212_PaleyI; 2026-10-03_Band1_Hadamard_12204_PaleyI; 2026-10-03_Band1_Hadamard_12164_PaleyI; 2026-10-03_Band1_Hadamard_12144_PaleyI; 2026-10-03_Band1_Hadamard_12120_PaleyI; 2026-10-03_Band1_Hadamard_12108_PaleyI; 2026-10-03_Band1_Hadamard_12072_PaleyI; 2026-10-03_Band1_Hadamard_12044_PaleyI; 2026-10-03_Band1_Hadamard_12012_PaleyI; 2026-10-03_Band1_Hadamard_12008_PaleyI; 2026-10-03_Band1_Hadamard_11988_PaleyI; 2026-10-03_Band1_Hadamard_11972_PaleyI; 2026-10-03_Band1_Hadamard_11960_PaleyI; 2026-10-03_Band1_Hadamard_11940_PaleyI; 2026-10-03_Band1_Hadamard_11928_PaleyI; 2026-10-02_Band1_Hadamard_11924_PaleyI; 2026-10-02_Band1_Hadamard_11904_PaleyI; 2026-10-02_Band1_Hadamard_11888_PaleyI; 2026-10-02_Band1_Hadamard_11868_PaleyI; 2026-10-02_Band1_Hadamard_11864_PaleyI; 2026-10-02_Band1_Hadamard_11840_PaleyI; 2026-10-02_Band1_Hadamard_11832_PaleyI; 2026-10-02_Band1_Hadamard_11828_PaleyI; 2026-10-02_Band1_Hadamard_11808_PaleyI; 2026-10-02_Band1_Hadamard_11784_PaleyI; 2026-10-02_Band1_Hadamard_11780_PaleyI; 2026-10-02_Band1_Hadamard_11744_PaleyI; 2026-10-02_Band1_Hadamard_11732_PaleyI; 2026-10-02_Band1_Hadamard_11720_PaleyI; 2026-10-02_Band1_Hadamard_11700_PaleyI; 2026-09-30_Band1_Hadamard_11588_PaleyI; 2026-09-30_Band1_Hadamard_11580_PaleyI; 2026-09-30_Band1_Hadamard_11552_PaleyI; 2026-09-30_Band1_Hadamard_11528_PaleyI; 2026-09-30_Band1_Hadamard_11520_PaleyI; 2026-09-30_Band1_Hadamard_11504_PaleyI; 2026-09-30_Band1_Hadamard_11492_PaleyI; 2026-09-30_Band1_Hadamard_11484_PaleyI; 2026-09-30_Band1_Hadamard_11472_PaleyI; 2026-09-30_Band1_Hadamard_11468_PaleyI; 2026-09-30_Band1_Hadamard_11448_PaleyI; 2026-09-30_Band1_Hadamard_11444_PaleyI; 2026-09-30_Band1_Hadamard_11424_PaleyI; 2026-09-30_Band1_Hadamard_11412_PaleyI; 2026-09-30_Band1_Hadamard_11400_PaleyI; 2026-09-30_Band1_Hadamard_11384_PaleyI; 2026-09-30_Band1_Hadamard_11352_PaleyI;

2026-09-30_Band1_Hadamard_11312_PaleyI; 2026-09-30_Band1_Hadamard_11300_PaleyI;
2026-09-30_Band1_Hadamard_11288_PaleyI; 2026-09-30_Band1_Hadamard_11280_PaleyI;
2026-09-30_Band1_Hadamard_11252_PaleyI; 2026-09-29_Band1_Hadamard_11244_PaleyI;
2026-09-29_Band1_Hadamard_11240_PaleyI; 2026-09-29_Band1_Hadamard_11172_PaleyI;
2026-09-29_Band1_Hadamard_11160_PaleyI; 2026-09-29_Band1_Hadamard_11132_PaleyI;
2026-09-29_Band1_Hadamard_11120_PaleyI; 2026-09-29_Band1_Hadamard_11088_PaleyI;
2026-09-29_Band1_Hadamard_11084_PaleyI; 2026-09-29_Band1_Hadamard_11072_PaleyI;
2026-09-29_Band1_Hadamard_11060_PaleyI; 2026-09-29_Band1_Hadamard_11048_PaleyI;
2026-09-29_Band1_Hadamard_11028_PaleyI; 2026-09-29_Band1_Hadamard_11004_PaleyI;
2026-09-29_Band1_Hadamard_10988_PaleyI; 2026-09-29_Band1_Hadamard_10980_PaleyI;
2026-09-29_Band1_Hadamard_10940_PaleyI; 2026-09-29_Band1_Hadamard_10892_PaleyI;
2026-09-29_Band1_Hadamard_10884_PaleyI; 2026-09-29_Band1_Hadamard_10860_PaleyI;
2026-09-29_Band1_Hadamard_10848_PaleyI; 2026-09-29_Band1_Hadamard_10832_PaleyI;
2026-09-29_Band1_Hadamard_10800_PaleyI; 2026-09-29_Band1_Hadamard_10772_PaleyI;
2026-09-29_Band1_Hadamard_10740_PaleyI; 2026-09-29_Band1_Hadamard_10724_PaleyI;
2026-09-29_Band1_Hadamard_10712_PaleyI; 2026-09-28_Band1_Hadamard_10692_PaleyI;
2026-09-28_Band1_Hadamard_10688_PaleyI; 2026-09-28_Band1_Hadamard_10668_PaleyI;
2026-09-28_Band1_Hadamard_10652_PaleyI; 2026-09-28_Band1_Hadamard_10640_PaleyI;
2026-09-28_Band1_Hadamard_10632_PaleyI; 2026-09-28_Band1_Hadamard_10628_PaleyI;
2026-09-28_Band1_Hadamard_10608_PaleyI; 2026-09-28_Band1_Hadamard_10568_PaleyI;
2026-09-28_Band1_Hadamard_10560_PaleyI; 2026-09-28_Band1_Hadamard_10532_PaleyI;
2026-09-28_Band1_Hadamard_10500_PaleyI; 2026-09-28_Band1_Hadamard_10488_PaleyI;
2026-09-28_Band1_Hadamard_10464_PaleyI; 2026-09-28_Band1_Hadamard_10460_PaleyI;
2026-09-28_Band1_Hadamard_10428_PaleyI; 2026-09-28_Band1_Hadamard_10400_PaleyI;
2026-09-24_Band1_Hadamard_10392_PaleyI; 2026-09-24_Band1_Hadamard_10344_PaleyI;
2026-09-24_Band1_Hadamard_10332_PaleyI; 2026-09-24_Band1_Hadamard_10304_PaleyI;
2026-09-24_Band1_Hadamard_10272_PaleyI; 2026-09-23_Band1_Hadamard_10268_PaleyI;
2026-09-23_Band1_Hadamard_10224_PaleyI; 2026-09-23_Band1_Hadamard_10212_PaleyI;
2026-09-23_Band1_Hadamard_10164_PaleyI; 2026-09-23_Band1_Hadamard_10160_PaleyI;
2026-09-23_Band1_Hadamard_10152_PaleyI; 2026-09-23_Band1_Hadamard_10140_PaleyI;
2026-09-23_Band1_Hadamard_10112_PaleyI; 2026-09-23_Band1_Hadamard_10104_PaleyI;
2026-09-23_Band1_Hadamard_10100_PaleyI; 2026-09-23_Band1_Hadamard_10068_PaleyI.

Lab. `/workspace/experiments/band1/hadamard_12240_paleyI/`.

Cutoff. 2026-10-03T23:04:06Z (~7:04PM ET).

Honesty stamp (non-negotiable).

This memo claims **finite construction of order 12240 only (verified max|HHˆT-nI|=0), via classical Paley-I q=12239 (prime ===3 mod 4). 12240 IN SPK via Paley-I.** Honest skip: Miyamoto digression this wave (prefer classical SPK). Classical SPK for 12240: Paley-I **PASS** ($q = 12239$ prime $\equiv 3 \pmod 4$); Paley-II **FAIL/SKIP** (q=6119=29*211 NOT prime; ===3 mod 4 needs ===1); Kronecker **FAIL/SKIP** (note $12240 = 4 \cdot 3060$; $3060 \equiv 0 \pmod 4$ order-admissible; factors $2^2 \cdot 3^2 \cdot 5 \cdot 17$; not_in_seeds / no H_3060). **12240**

folds into SPK **only** via classical Paley-I (this run). It **does not** claim the full Hadamard conjecture. GATE-04 collection remains **LOCKED**. No RH / P≠NP / YM / Frankl / NS-solved / AC-solved claims. Gap **2052** remains **OPEN**. `clay_closed = false`. Prior SPK **12228+12212+12204+12164+12144+12120+12108+12072+12044+12012+12008+11988+11972+119** retained; Miyamoto-only **8948/8644/8516/8452/7988/...** excluded.

ewpage

Abstract

Finite construction of order **12240**.

- H_{12240} = Paley-I ($q = 12239$) – classical SPK
- Paley-II for 12240: **FAIL/SKIP** ($q = 6119 = 29 \cdot 211$ NOT prime; $\equiv 3 \pmod 4$ needs $\equiv 1$)
- Kronecker for 12240: **FAIL/SKIP** ($4 \cdot 3060$; $3060 \equiv 0 \pmod 4$ order-admissible; factors $2^2 \cdot 3^2 \cdot 5 \cdot 17$; not_in_seeds / no H_3060)
- Skip: Miyamoto digression this wave

SPK: IN SPK via Paley-I.
Gap 2052: OPEN. **Conjecture:** OPEN. **clay_closed:** false. **GATE-04:** LOCKED.

`python3 verify_hadamard_12240.py`

Results. overall_pass=True; units 441/441; row blocks 96/96 (block size 128); wall 1547.5592s; constructed max|HHˆT-nI|=0.

Cutoff: 2026-10-03T23:04:06Z (~7:04PM ET)

Arithmetic

- Target $n = 12240 = q + 1$ with $q = 12239$ prime $\equiv 3 \pmod 4$ — Paley-I eligible.
- Paley-II: $q = 6119 = 29 \cdot 211$ NOT prime; $\equiv 3 \pmod 4$ needs $\equiv 1$ — FAIL/SKIP.
- Kronecker: $12240 = 4 \cdot 3060$; $3060 \equiv 0 \pmod 4$ order-admissible; factors $2^2 \cdot 3^2 \cdot 5 \cdot 17$; not_in_seeds / no H_3060 — FAIL/SKIP.
- Miyamoto digression skipped this wave (prefer classical SPK).

Constructions (verified)

Object	Method	max|HHˆT-nI|	Wall (s)
H_12240 Paley-I	Paley-I q=12239	0	1520.0716

Object	Method	max|HHˆT-nI|	Wall (s)
H_12240 Paley-II	FAIL/SKIP (q=6119=29*211 NOT prime; ===3 mod 4 needs ===1)	–	–
H_12240 Kronecker	FAIL/SKIP (4*3060; 3060===0 mod 4 order-admissible; factors 2ˆ23ˆ2517; not_in_seeds / no H_3060)	–	–

SPK closure note

Seeds include prior SPK through **11528** lineage (**11528**; **11520**; **11504**; **11492**; **11484**; **11472**; **11468**; **11448**; **11444**; **11424**; **11412**; **11400**; **11384**; **11352**; **11312**; **11300**; **11288**; **11280**; **11252**; **11244**; **11240**; **11172**; **11160**; **11132**; **11120**; **11088**; **11084**; **11072**; **11060**; **11048**; **11028**; **11004**; **10988**; **10980**; **10940**; **10904**; **10892**; **10884**; **10860**; **10848**; **10832**; **10800**; **10772**; **10740**; **10724**; **10712**; **10692**; **10688**; **10668**; **10652**; **10640**; **10632**; **10628**; **10608**; **10568**; **10560**; **10532**; **10500**; **10488**; **10464**; **10460**; **10428**; **10400**; **10392**; **10344**; **10332**; **10304**; **10272**; **10268**; **10260**; **10248**; **10244**; **10224**; **10212**; **10164**; **10160**; **10152**; **10140**; **10112**; **10104**; **10100**; **10068**; **10040**; **10008**; **9968**; **9932**; **9924**; **9908**; **9888**; **9884**; **9872**; **9860**; **9852**; **9840**; **9812**; **9804**; **9792**; **9788**; **9768**; **9744**; **9740**; **9720**; **9680**; **9644**; **9632**; **9624**; **9620**; **9588**; **9552**; **9548**; **9540**; **9512**; **9492**; **9480**; **9468**; **9420**; **9404**; **9392**; **9372**; **9344**; **9324**; **9320**; **9312**; **9284**; **9240**; **9228**; **9204**; **9200**; **9188**; **9152**; **9104**; **9128**; **9124**; **9092**; **2220+8884**; **2236**; **2160**; **2140+8564**; …; **NOT** 8948/8644/8516/…) + verified **10464** (+ prior **10392+10344+10332+10304+10272+10268+10224+10212+10164+10160+10152+10140+10112+101**
Next constructive gap **2052** (OPEN).

Quantity	Value
11520 in SPK	**True** (Paley-I q=12239)
Gap 2052	**OPEN**
clay_closed	false
GATE-04	LOCKED

Honesty

- clay_closed = false
- GATE-04 LOCKED
- Hadamard conjecture OPEN
- Gap 2052 OPEN (method triage / non-finding only)
- No RH / P!=NP / YM / Frankl / NS / AC solved claims
- SPK membership only via classical Paley-I/II/Kronecker (not Miyamoto alone)

Next optional

q=12251->n=12252 Paley-I DIRECT (or gap 2052 / Track B)